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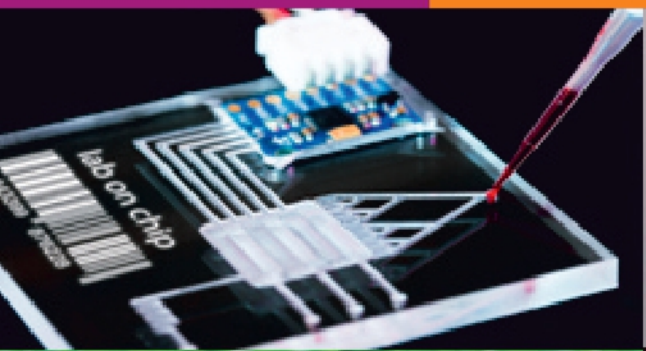
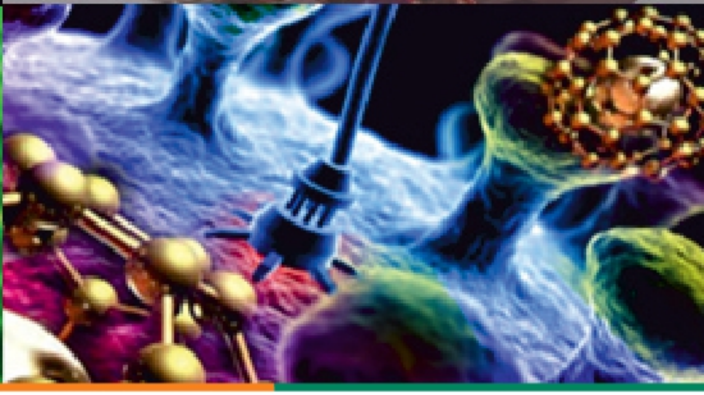
ABSTRACTS







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PP-010

Repurposing the Dark Genome – I: Antisense Proteins

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Abstract: Based on the expression patterns, genomes are divided into three categories: sequences that encode proteins, sequences that encode RNA, and sequences that don't participate in expression processes. Recent sequencing and annotation have confirmed that a minor portion of the genome is tasked with protein encoding, while a significant section encodes RNA, and the remaining portion of DNA sequences does not actively contribute to the expression of genes. This allocation ratio varies among different organisms. Is it feasible to create proteins and peptides synthetically using non-expressive sequences? This study extends the prior research conducted by Dhar et al. in 2009, which involved creating intergenic proteins in E. coli. It explores the possibilities of making functional genes and proteins from antisense DNA sequences. We computationally translated antisense strands of 4315 protein-coding genes in *E. coli* and 6317 genes in *S. cerevisiae*, producing 32 and 10 full-length proteins, respectively. Among these, 9 in *E. coli* and 7 in *S. cerevisiae* were found to be unique. Furthermore, antisense proteins exhibited isoelectric points, instability indexes, and hydropathy values in the promising range, suggesting potential structural stability if these proteins were expressed within cells. Many of the antisense proteins indicated strong possibilities of transporter and enzyme functions. Currently the experimental validation studies are going on. Designing novel antisense genes, RNA, and proteins/peptides from the dark genome points to a huge untapped space that may yield a wealth of information from cell physiology, evolutionary, and application perspectives.

Key Words: Antisense protein, non-expressing, protein-coding.

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